

Spatiotemporal Filtering Bank for Decoding Motor Imagery EEG Signals

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Abstract—Common Spatial Patterns (CSP) have been extensively utilized in motor imagery brain-computer interfaces (BCIs). However, the frequency band and spatial specificity of subjects during motor imagery have limited the further development of CSP. To overcome these challenges, this paper introduces a novel algorithm called spatiotemporal filtering bank and classification (STFBC). Specifically, inspired by the concept of Filter Bank Common Spatial Patterns (FBCSP), we developed a spatiotemporal filter bank to extract fine-grained spatiotemporal features from EEG data. Subsequently, a heuristic feature selection method was employed to automatically identify a subset of features with low redundancy, high correlation, and complementarity for classification. To validate the effectiveness of the proposed algorithm, we conducted experiments using two public datasets: BCI Competition III dataset IVa and BCI Competition IV dataset IIa. In comparison to other competitive algorithms, the STFBC algorithm achieved the highest classification accuracy on these two datasets, with rates of 86.67% and 81.69%, respectively. Furthermore, the spatial activation patterns revealed by STFBC were consistent with the brain regions activated during the motor imagery task.

Keywords—component; motor Imagery; spatiotemporal filter; common spatial patterns; feature Selection; electroencephalogram

I. INTRODUCTION

Brain-computer interface (BCI) technology involves the acquisition of electroencephalogram (EEG) signals that record the activity of brain neurons, and employs signal processing and machine learning methodologies to decode the variations in brain activity. This process facilitates direct communication and control between the human brain and external devices [1], [2], [3], [4]. Among the various research directions currently being explored, BCI technology that utilizes the motor imagery paradigm has become a primary focus. This field emphasizes the effective extraction of EEG signal features and the accuracy of classification. Previous studies [5], [6] have indicated that during motor imagery, the sensory-motor areas of the brain exhibit event-related synchronization (ERS) and event-related desynchronization (ERD) phenomena. The CSP method is capable of effectively capturing these phenomena, thereby extracting discriminative features, which has led to its widespread application in motor imagery EEG decoding [7], [8].

The CSP algorithm aims to identify a set of optimal spatial filters using matrix diagonalization [9]. These filters can effectively project the original EEG signal into a potential space

and extract a discrimination feature vector in that space that can maximize the variance ratio between the two types of signals. However, the traditional CSP method still has some limitations in noise sensitivity and multi-channel analysis. However, traditional CSP methods exhibit certain limitations in terms of noise sensitivity and multi-channel analysis. To enhance the robustness of the covariance matrix in scenarios with small sample sizes, Lotte et al. [10] proposed the regularized CSP (RCSP) method. This approach incorporates regularization terms into the covariance matrix to adjust the diagonal elements, thereby improving the algorithm's robustness. To address the specificity of response frequency and response time during motor imagery tasks, researchers have employed a sliding time window technique to select the optimal time window. Additionally, Ang et al. introduced the FBCSP algorithm to tackle the issue of frequency band selection within the CSP framework [11]. This algorithm utilizes a bank of bandpass filters to decompose the signals, extracting CSP features from each frequency band and employing mutual information for feature selection.

In the aforementioned CSP-based methods, spatial filtering and temporal filtering are typically performed in a sequential manner, starting with bandpass filtering followed by spatial filtering. However, this two-step approach overlooks the inherent temporal response and spatial topology of the EEG signals, which may result in suboptimal decoding performance. To address this limitation, some studies have improved the CSP algorithm to simultaneously optimize the spatiotemporal filters [12], [13], [14]. For instance, RSTFC provides a unified framework for maximizing the variance ratio to optimize spatiotemporal filters, employing an L2-norm regularization technique for supervised spatiotemporal filtering [15]. Similarly, STECS [12] introduces group sparsity constraints, integrating the channel selection problem into the spatiotemporal filter optimization framework. SBLEST [16] jointly optimizes spatiotemporal filters and classifiers within the sparse Bayesian learning framework, thereby enhancing decoding accuracy. However, the fixed order of the temporal filter and the predetermined time delay may lead to the selection of frequency bands that are not directly related to motor imagery tasks. This approach lacks the flexibility required to accommodate the unique needs of different subjects, which may consequently impact the overall effectiveness of the decoding process.

To tackle these issues, this paper proposes a novel framework called spatiotemporal filtering bank and

classification (STFBC). these issues, this paper proposes a novel framework called spatiotemporal filtering bank and classification (STFBC). This framework builds upon the concept of multi-frequency bands from FBCSP by constructing a bank of spatiotemporal filters to extract more granular spatiotemporal features. Furthermore, due to the volume conduction effects present in the human head, the electrical activity measured at the sensor level is a mixture of neural signals from multiple potential sources. Consequently, the extracted spatiotemporal features may exhibit high redundancy or similar information, which increases the risk of overfitting when training classifiers. To mitigate this issue, we employ a heuristic feature selection (FS) algorithm that automatically identifies a subset of features characterized by low redundancy, high similarity, and complementarity from a pool of candidate spatiotemporal features. These selected features correspond to the optimal spatial and temporal filters tailored to each subject, thereby enhancing the model's decoding performance. The overall architecture of STFBC is illustrated in Fig. 1.

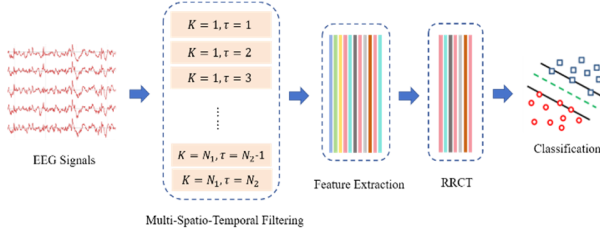


Figure 1. Framework of the STFBC method

II. MATERIALS AND METHODS

A. Datasets

Dataset IVa from BCI Competition III (Dataset 1) recorded the motor imagination tasks of five subjects (aa, al, av, aw, ay) with a 118-channel electrode cap. Each subject participated in three types of tasks: imagining the left hand, right hand, and right foot, with each task lasting 3.5 seconds. The total number of experimental trials for each subject was 280, with the training sample sizes for subjects aa, al, av, aw, and ay being 168, 224, 84, 56, and 28, respectively; the remaining samples were designated for testing. For more detailed information, please refer to reference [17]. Since this dataset provides only right-hand and right-footed samples, we focused our classification efforts solely on these two task categories.

Dataset IIa from BCI Competition IV [18] (Dataset 2) recorded EEG signals from nine healthy subjects (A01-A09) using 22 electrodes while they performed four types of motor imagery tasks, labeled as left hand, right hand, both feet, and tongue. The signal was sampled at a rate of 250 Hz. The experiment was divided into two sessions (T and E), with each subject participating in a total of 288 trials per session, and each task being repeated 72 times. For binary classification, each subject generates $C_4^2 = 6$ class pairs, resulting in a total of $6 \times 9 = 54$ data subsets.

B. Preprocessing

For Dataset 1, we followed the recommendations from the literature [19] and utilized 49 channels for our analysis. In

contrast, for Dataset 2, we employed all 22 channels. To retain the EEG signals within the target frequency band, we used a 6-order Butterworth bandpass filter to perform 7-30 Hz bandpass filtering on the raw experimental data of the two datasets. Following the filtering process, we extracted the time segment from 0.5 to 3.5 seconds, where 0 seconds corresponds to the moment the cue ends.

C. Feature Extraction

Let $X \in R^{C \times T}$ represent a single trial from recorded EEG signals, where C denotes the number of channels (i.e., electrode leads) and T represents the number of time samples for each channel. First, we construct a channel finite impulse response (FIR) temporal filter of order K to extract the spectral band signals for each channel. The definition of the temporal filter $f_c(t) \in R^{T \times 1}$ is as follows:

$$f_c(t) = b_{c1}x_c(t) + b_{c2}x_c(t - \tau) + \dots + b_{cK}x_c(t - (K - 1)\tau), \quad (1)$$

where $b_c = [b_{c1}, b_{c2}, \dots, b_{cK}]^T \in R^{K \times 1}$ ($c \in \{1, 2, \dots, C\}$) is the FIR filter associated with c -th channel, $t = 1, 2, \dots, T$ and τ represents the delay time factor between adjacent input samples for each FIR filter.

For a given spatial filter $s = [s_1, s_1, \dots, s_C]^T \in R^{C \times 1}$, applying this filter to the EEG signals that have undergone temporal filtering yields the extraction of the latent signal $z \in R^{T \times 1}$ after spatial filtering.

$$z = [f_1, f_2, \dots, f_C] \cdot s. \quad (2)$$

Substituting (1) into (2) and applying a linear transformation yields the following more concise expression.

$$z = \tilde{X}^T \cdot w, \quad (3)$$

where w is the spatiotemporal filter which needs to be optimize and has the follows forms:

$$w = [s_1 b_{11}, s_2 b_{21}, \dots, s_C b_{C1}, \dots, s_1 b_{1K}, s_2 b_{2K}, \dots, s_C b_{CK}]^T = [w_{11}, w_{21}, \dots, w_{C1}, \dots, w_{1K}, w_{2K}, \dots, w_{CK}]^T, \quad (4)$$

where $\tilde{X} = [X^T, [X^{(\tau)}]^T, \dots, [X^{((K-1)\tau)}]^T]^T$ is an augmented data matrix in which $[X^{(\tau)}] \in R^{C \times T}$ is the original data X delayed by τ sampled time points and $B = [b_1, b_2, \dots, b_C]$.

By examining (3), a natural approach is to utilize CSP to derive the filter w from $\tilde{X}$, and subsequently decompose w into a spatial filter s and a temporal filter f_c . Specifically, let Σ_1 and Σ_2 represent the average spatial covariance matrices for the two conditions calculated from $\tilde{X}$. The filter w can be computed as follows:

$$w = \underset{w}{\operatorname{argmax}} \frac{w^T \Sigma_1 w}{w^T \Sigma_2 w}. \quad (5)$$

The above Rayleigh quotient can be converted to a generalized eigenvalue problem.

$$\Sigma_1 w = \lambda \Sigma_2 w, \quad (6)$$

where λ and w are the eigenvalues and eigenvector of the generalized eigenvalue equation. According to [15], the spatial filter s and temporal filter f can be computed as:

$$s = [s_1, s_2, \dots, s_C]^T, \quad (7)$$

where $s_c = \text{sign}[w_{c1}] \cdot \sqrt{(w_{c1})^2 + (w_{c2})^2 + \dots + (w_{cK})^2}$ and $\text{sign}[x]=1$ when $x \geq 0$, and $\text{sign}[x]=-1$ otherwise.

$$f_c = \frac{1}{s_c} \cdot [w_{c1}, w_{c2}, \dots, w_{cK}]^T, c \in \{1, 2, \dots, C\}. \quad (8)$$

To capture ERD and ERS during motor imagery tasks, we utilize the logarithmic variance of EEG signal after applying spatiotemporal filtering as the features. Building on previous research, we stack the eigenvectors corresponding to the M largest/smallest eigenvalues to construct a spatiotemporal filter $W = [w_1, \dots, w_{2M}]$, where $M = 3$ in this study. Subsequently, we applied W to the augmented data matrix $\tilde{X}$ for spatial filtering and extracted the CSP spatial features as follows:

$$z = [z_1, z_2, \dots, z_{2M}]^T, \quad (9)$$

where $z_m = \log(\text{var}(w_m^T \tilde{X}))$, $m = 1, \dots, 2M$. It is important to note that for each value of τ , a total of $2M$ features can be extracted. Specifically, when considering $\tau \in \{1, 2, \dots, N_2\}$, the total number of features that can be obtained is $2N_2M$.

D. Feature Selection

From (1), it can be concluded that when τ and K take on different values, they correspond to distinct FIR filters. Similar to the concept of FBCSP, applying these filters to the augmented data $\tilde{X}$ generates multiple spatiotemporal features. However, the features of higher dimensions will bring redundant information and hinder the generalization ability of the classification model. To improve the characterization of features and simplify the subsequent classification model, we employ a feature selection method known as relevance, redundancy, and complementarity trade-off (RRCT) [20]. This heuristic feature selection algorithm simultaneously considers the correlation, redundancy, and completeness among features, thereby selecting a robust subset of features to enhance the model's generalization performance.

E. Classification

After feature extraction, we employed two commonly used supervised classification methods, such as linear discriminant Analysis (LDA) and support vector machine (SVM).

III. EXPERIMENTAL RESULTES

In this section, we conduct a subject-dependent binary classification experiment on two datasets to evaluate the classification performance of the proposed method.

A. Experimental Protocol

To validate the effectiveness of the proposed algorithm, we compared its performance with that of other competing algorithms across two datasets. The relevant algorithms and their parameter configurations are as follows:

a) **CSP** [7]: CSP method is employed to extract spatial features from the preprocessed data.

b) **FBCSP** [11]: First, six sub-band components (e.g., 7-11 Hz, 11-15 Hz, 15-19 Hz, 19-23 Hz, 23-27 Hz, and 27-30 Hz) were extracted from raw EEG signals using six 6th-order Butterworth filters. Subsequently, CSP feature extraction was performed on each component [13], followed by the selection of optimal individual features based on a mutual information strategy.

c) **CSSSP** [21]: The common spatial-spectral pattern with sequential processing (CSSSP) improves upon CSP by removing the constraints on the order of FIR filters. The regularization parameter t was turned from $\{0.1, 0.2, \dots, 5\}$ and the order of FIR filter was set to $K = 6$.

d) **SBLEST** [16]: Within the framework of principled sparse Bayesian learning, this method simultaneously optimizes the spatiotemporal filters and classifiers to maximize predictive accuracy. The parameter delay time τ was turned from $\{0, 1, 2, 3, 4, 5\}$ and the order of FIR filter was set to $K = 1$ when $\tau = 0$ and $K = 2$ otherwise.

For the CSP, FBCSP and CSSSP algorithms, fisher linear discriminant analysis (FLDA) is employed for classification following the extraction of spatial or spatiotemporal features. In our STFBC model, we employed SVM (using the 'fitsvm' function) and LDA (using the 'fitdiscr' function) in MATLAB to classify the spatiotemporally filtered EEG data. For the SVM, we selected the radial basis function (RBF) as the kernel function while keeping the other parameters at their default settings. To assess the stability and reliability of models, a 10-fold cross-validation strategy was employed to evaluate the generalization ability of each method across two datasets. Additionally, the number of CSP filters in our method was set to 6 (i.e., $M = 3$).

B. Results and Discussion

Table I presents the average classification accuracy of all algorithms across the two datasets. As shown in Table I, the proposed STFBC model achieves the highest classification performance compared to the other five competitive algorithms, with significant differences observed ($p < 0.05$, Wilcoxon signed-rank test). Specifically, STFBC attained the average classification accuracies of 86.67% and 81.69% on datasets 1 and 2, respectively, representing improvements of 2.53% and 2.86% over the SBLEST model. In comparison to SBLEST, we adopted the concept of FBCSP by extending the spatiotemporal filter into a group of spatiotemporal filters to capture more fine-grained spatiotemporal features of the subjects. The experimental results demonstrate the practicality and effectiveness of the proposed method. Furthermore, the average classification accuracy of CSSSP, SBLEST and STFBC surpasses than that of CSP. This finding indicates that spatiotemporal filtering reveals more decoding information than spatial filtering, which is essential for understanding the specific time-frequency and spatial responses of individuals during motor imagery tasks.

TABLE I. MEAN CLASSIFICATION ACCURACIES (%) OF CSP, FBCSP, CSSSP, SBLES AND STFBC

Dataset	CSP	FBCSP	CSSSP	SBLES T	STFBC +SVM	STFBC +LDA
Dataset 1	77.45±1 3.94	72.52±1 4.61	78.66±1 4.95	84.14±1 1.61	86.67±8 .77	85.86±9 .38
Dataset 2	77.07±1 3.70	78.99±1 2.25	77.74±1 3.58	78.83±1 2.19	81.33±1 3.62	81.69±1 3.54

C. Classification Performance with Different Number of Features

In our model, we employed a recently proposed heuristic feature selection method known as RRCT. This method selects a subset of features from the candidate set that exhibits low redundancy, high correlation, and complementarity. Such characteristics are particularly beneficial for decoding tasks involving EEG signals, which are often characterized by high redundancy and low signal-to-noise ratios.

Fig. 2 illustrates the average accuracy of the SVM classifier as a function of the number of features. It is evident that when the number of features is relatively low (for example, between 4 and 6 features), our model achieves optimal classification performance. However, as the number of features increases, there is a slight decline in model performance. This phenomenon may be attributed to the introduction of redundant information due to an excessive number of features, which hinders the model's decoding capability. Therefore, the judicious selection of feature quantity is crucial for enhancing model performance; an overabundance of features may lead to information confusion, adversely affecting classification outcomes. By utilizing the RRCT method, we can effectively balance the quantity and quality of features, thereby optimizing the overall performance of the model.

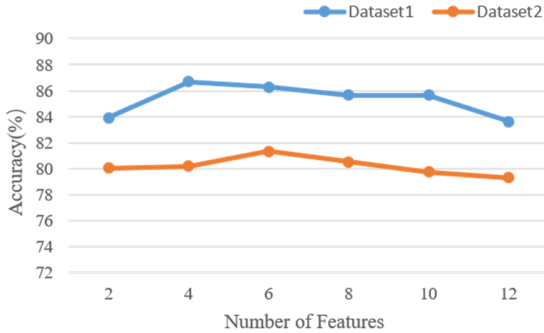


Figure 2. Variation of classification accuracy with the number of features

D. Visualization

Fig. 3 illustrates the topographic maps of spatial filters for representative subjects across all binary classification experiments conducted on the two datasets. The classification performance of these representative subjects in each case is presented in Table II. The analysis of motor imagery for the representative subjects, as shown in Fig. 3, indicates that the scalp maps obtained through the model are primarily concentrated in regions associated with the task-related areas of

the motor cortex, which aligns with existing research findings [22], [23], [24]. Specifically, the imagery of the left hand is controlled by the motor cortex in the right hemisphere of the brain, while the imagery of the right hand is predominantly governed by the motor cortex in the left hemisphere [25], [26]. Furthermore, the involvement of the tongue and feet in more complex motor imagery or perceptual tasks often results in more pronounced cortical activity, which may enhance the discriminability of classification.

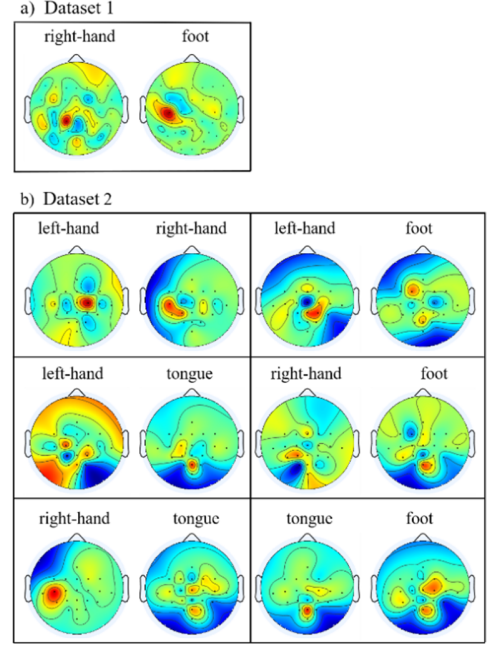


Figure 3. Scalp maps of the selected spatial filters of subject aa in dataset 1 (a) and A01 in dataset 2 (b) for all binary classification experiments

TABLE II. CLASSIFICATION ACCURACIES (%) OF SUBJECT AA IN DATASET 1 AND SUBJECT A01 IN DATASET 2 FOR ALL BINARY CLASSIFICATION EXPERIMENTS USING SVM CLASSIFIER

Subject	R vs F					
aa from Dataset 1	87.57					
Subject	L vs R	L vs F	T vs L	F vs R	T vs R	T vs F
A01 from Dataset 2	83.24	72.14	70.14	76.95	93.76	92.24

R stands for the left hand, L stands for the right hand, T stands for the tongue, and F stands for the feet

IV. CONCLUSION

In this study, we present a novel EEG decoding algorithm based on a group of spatiotemporal filters, referred to as STFBC. STFBC integrates multiple spatiotemporal filters with a heuristic feature selection method to identify a subset of features that exhibit low redundancy, high correlation, and complementarity, thereby enhancing decoding capability. We assessed the proposed model on two publicly available motor imagery datasets. The results reveal that the STFBC algorithm outperforms other competing algorithms, highlighting our approach as a promising framework for accurately decoding

motor imagery patterns and advancing the development of high-performance brain-computer interfaces.

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